Jiwan Chung

Postdoctoral Researcher, Yonsei University

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AI researcher studying how multimodal models understand, reason, and interact, and improving these capabilities through changes to model architectures and training objectives, guided by fine-grained evaluation.

Selected Research

Distance-aware training for discrete outputs ICLR 2026 | First author

Teaching Metric Distance to Discrete Autoregressive Language Models

- Developed DIST^{2}Loss, a training objective that turns distances between numeric, spatial, or quantized tokens into soft supervision for discrete autoregressive models.
- Improved performance over cross-entropy-only training in visual grounding, robotic manipulation, reward modeling, and image generation, with larger gains under limited data.

Why recurrent models lose track of state NeurIPS 2026 | First author

Rethinking State Tracking in Recurrent Models Through Error Control Dynamics

- Showed why SSMs and linear attention can fail despite sufficient expressive power: preserving distinct states exactly can leave some errors uncorrected.
- Predicted when trained state trackers fail on longer sequences by measuring how accumulated errors make different states indistinguishable.

Diagnosing multimodal reasoning NeurIPS 2026 | First author

What MLLMs Learn When They Learn Multimodal Reasoning

- Built MathLens, a formally validated framework for decomposing multimodal reasoning capability into perception, reasoning, and their interaction.
- Found that gains in the evaluated RL-trained models primarily reflected better perception and diagram robustness, while textual SFT supported reflective reasoning.

Research Experience

Yonsei University, CIP Lab Sep 2026 - Present

Postdoctoral Researcher

Microsoft Research, AI Frontiers May 2025 - Aug 2025

Research Intern

Developed MathLens during the internship. Continued collaborating with the team on process-level evaluation of web agents through WebStep.

LG AI Research, AML Mar 2024 - Jul 2024

Research Intern

Developed DIST^{2}Loss for autoregressive models predicting coordinates, robotic actions, and image tokens.

Naver Cloud, Foundational Research Jul 2023 - Jan 2024

Research Intern

Contributed to HyperCLOVA X development and explored omnimodal language modeling.

Education

Yonsei University 2023 - 2026
Ph.D. in Artificial Intelligence

Seoul National University 2019 - 2023
M.S. in Computer Engineering

Yonsei University 2014 - 2018
B.Sc. in Computer Science and Philosophy (double major)

Additional Selected Publications

* Equal contribution. Full publication record: Google Scholar.

Global Geometry Is Not Enough for Vision Representations
J. Chung and S. J. Kim | **ICML 2026**

v1: Learning to Point Visual Tokens for Multimodal Grounded Reasoning
J. Chung*, J. Kim*, S. Kim, J. Lee, M. S. Kim, and Y. Yu | **COLM 2026**

Where Did It Go Wrong? Process-Level Evaluation of Web Agents with Semantic State Tracking
J. Chung, J. Byun, V. Vineet, and S. J. Kim | **COLM 2026**

Are Any-to-Any Models More Consistent Across Modality Transfers Than Specialists?
J. Chung, J. Yoon, J. Park, et al. | **ACL 2025**

MASS: Overcoming Language Bias in Image-Text Matching
J. Chung, S. Lim, S. Lee, and Y. Yu | **AAAI 2025**

Speaking Beyond Language: A Large-Scale Multimodal Dataset for Learning Nonverbal Cues from Video-Grounded Dialogues
J. Chung*, Y. Kim*, et al. | **ACL 2025**

Selective Vision Is the Challenge for Visual Reasoning: A Benchmark for Visual Argument Understanding
J. Chung*, S. Lee*, M. Kim, et al. | **EMNLP 2024, Oral**

Fusing Pre-Trained Language Models with Multimodal Prompts through Reinforcement Learning
J. Chung*, Y. Yu*, H. Yun, et al. | **CVPR 2023**

VLIS: Unimodal Language Models Guide Multimodal Language Generation
J. Chung and Y. Yu | **EMNLP 2023**

ACAV100M: Automatic Curation of Large-Scale Datasets for Audio-Visual Video Representation Learning
J. Chung*, S. Lee*, Y. Yu, et al. | **ICCV 2021**

Service and Honors

Top Reviewer, NeurIPS 2026; Gold Reviewer, ICML 2026.

Reviewer: NeurIPS, ICLR, ICML, COLM, CVPR, ICCV, ACL Rolling Review, TPAMI, and TMLR.

Selected awards: AI Seoul Tech Graduate Scholarship (2025); first place, DramaQA Challenge (ECCV Workshop, 2020) and LSMDC Challenge (ICCV Workshop, 2019).